

Gemmini Accelerator Backend

Reference Manual

ONNX-MLIR Integration & Implementation Guide

Version 3.1 (Corrected)

SST Team

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Abstract: This reference manual provides comprehensive documentation for the Gemmini accelerator backend within the ONNX-MLIR compiler framework. The backend enables ONNX models to target the Gemmini systolic array accelerator on RISC-V platforms. As of June 2026, the backend passes 41/41 MLIR lit tests, 18/18 integration tests, 22/22 example tests, and 9/9 model zoo models. This document covers architecture, compiler passes, MLIR dialects, runtime library, build configuration, testing infrastructure, validation results, usage examples, and troubleshooting.

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1 Introduction

1.1 Document Purpose and Scope

This reference manual serves as the authoritative guide for developers, researchers, and system architects working with the Gemmini accelerator backend in ONNX-MLIR. It documents the complete implementation, from high-level architecture to low-level runtime details.

1.2 Core Definitions

Definition: Gemmini

Gemmini is an open-source, configurable systolic array generator developed by UC Berkeley. It produces RISC-V RoCC (RISC-V Custom Coprocessor) accelerators for dense linear algebra operations, particularly matrix multiplication and convolution. The generator is written in Chisel and produces synthesizable Verilog for FPGA or ASIC implementations.

Key properties:

- Configurable systolic array dimensions (default 16x16)
- Support for weight-stationary (WS) and output-stationary (OS) dataflows
- Software-managed scratchpad memory
- DMA engine for data movement
- Optional activation functions (ReLU, ReLU6)

Definition: Systolic Array

A systolic array is a homogeneous network of processing elements (PEs) that rhythmically compute and pass data through the network. Gemmini's default configuration uses a 16x16 systolic array where each PE performs a multiply-accumulate (MAC) operation every cycle.

Characteristics:

- Data flows through the array in a rhythmic, pipelined fashion
- Each PE communicates only with its nearest neighbors
- Reduces memory bandwidth requirements by reusing data
- Achieves high throughput for matrix multiplication

Definition: RoCC (RISC-V Custom Coprocessor)

RoCC is a RISC-V extension that defines a standard interface for attaching custom coprocessors. Gemmini is implemented as a RoCC accelerator using the RoCC port of a Rocket or BOOM tile.

Instruction format:

```
.insn r CUSTOM_3, 0x3, funct7, x0, rs1, rs2
```

Definition: Scratchpad Memory

Software-managed on-chip SRAM, distinct from hardware-managed caches. Gemmini's scratchpad has 16,384 rows x 16 bytes per row (256 KB total) organized in 4 banks for parallel access. This is 16x larger than the stub values used in early development.

Definition: Double Buffering (Ping-Pong)

A technique that uses two memory buffers to overlap data transfer with computation. While the systolic array computes on one buffer, DMA loads the next tile into the other buffer. The slot offsets are expressed as relative row ranges within the 16,384-row address space.

Definition: Spike Simulator

The official RISC-V ISA simulator. With the `-extension=gemmini` flag, Spike loads a shared library (`libgemmini.so`) that emulates Gemmini RoCC instructions.

Definition: ONNX-MLIR

An open-source compiler that translates ONNX (Open Neural Network Exchange) models into MLIR (Multi-Level Intermediate Representation) and then to LLVM IR, ultimately producing executable binaries for various targets including RISC-V.

1.3 Current Status Summary

Test Suite	Count	Status
MLIR lit tests	41	41/41 PASS
Integration tests	18	18/18 PASS
Example tests	22	22/22 PASS
Model zoo	9	9/9 PASS
Numerical correctness (Spike)	3 models	PASS at 1e-5

Table 1: Current Validation Status

2 Quick Start Guide (5 Minutes)

2.1 Step 1: Check Prerequisites

```
1 scripts/check_prerequisites.sh --mode compile-only
```

2.2 Step 2: Build the Compiler

```
1 cmake -S . -B gemmini_toolchain_build -DONNX_MLIR_ENABLE_GEMMINI=ON -
   DONNX_MLIR_GEMMINI_MODE=compile-only -DCMAKE_BUILD_TYPE=Release
2 cmake --build gemmini_toolchain_build -j$(nproc) --target onnx-mlir
```

2.3 Step 3: Compile ResNet-18

```
1 ONNX_MLIR=gemmini_toolchain_build/Release/bin/onnx-mlir
2 $ONNX_MLIR --maccel=Gemmini --EmitMLIR resnet18-v1-7.onnx -o /tmp/
  resnet18_gemmini
```

2.4 Step 4: Count Gemmini Calls

```
1 grep -c 'om_gemmini_' /tmp/resnet18_gemmini.onnx.mlir
```

```
21
```

2.5 Step 5: Run on Spike (Simulator Mode)

```
1 RISCVC_INSTALL="$HOME/riscv-gemmini" scripts/setup_gemmini_simulator.sh
  --install
2 cmake -S . -B gemmini_toolchain_build_sim -DONNX_MLIR_ENABLE_GEMMINI=ON
  -DONNX_MLIR_GEMMINI_MODE=simulator -DRISCVC_INSTALL="$HOME/riscv-
  gemmini"
3 cmake --build gemmini_toolchain_build_sim -j$(nproc)
4 $ONNX_MLIR --maccel=Gemmini --mtriple=riscv64-unknown-linux-gnu
  resnet18-v1-7.onnx -o resnet18_rv64
5 spike --extension=gemmini --isa=rv64imafdc "$HOME/riscv-gemmini/bin/pk"
  ./resnet18_rv64
```

2.6 Expected Output

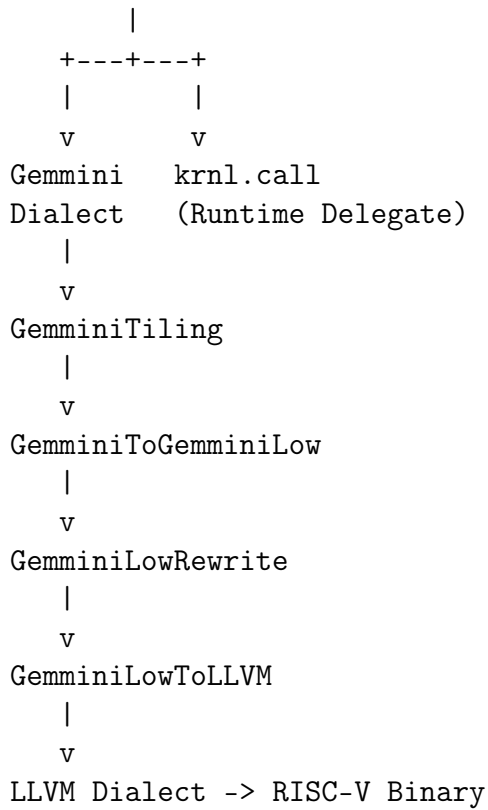
Item	Expected Value
Gemmini runtime calls	21 (20 om_gemmini_conv_f32 + 1 om_gemmini_gemm_f32_bias)
MLIR lit tests pass rate	41/41 (100%)
Integration tests pass rate	18/18 (100%)

Table 2: Expected Quick Start Results

3 Architecture Overview

3.1 Compilation Pipeline

```
ONNX Model (.onnx)
  |
  v
ONNX Dialect
  |
  v
ONNXToGemmini Pass (benefit=10)
```



3.2 Two Execution Paths

Aspect	Runtime Delegate Path	Direct RoCC Path
Primary ops	Conv, Gemm, MatMul f32, Re-size, Pad, Concat, Slice, Transpose, Split, Sigmoid, Mul, Relu, BN, Pooling, Softmax	MatMul int8/i32
Lowering	krnl.call @om_gemmini_*	gemmini_low.* to CUS-TOM_3 asm
Hardware use	Quantize f32 to int8, call tiled_matmul_auto, dequantize	Direct RoCC instructions
Benefit	10 (wins over standard path)	11 (highest priority)

Table 3: Two Execution Paths Comparison

3.3 Hardware Parameters

Parameter	Value	Description
Systolic array dimension	DIM = 16	16x16 processing elements
Scratchpad banks	4	Independent memory banks
Scratchpad rows	16,384	Total addressable rows (256 KB)
Accumulator rows	1,024	Accumulator buffer rows (64 KB)
Element type	int8	Input element type
Accumulator type	int32	Accumulator element type
RoCC channel	XCUSTOM_ACC = 3	CUSTOM_3 opcode (0x7b)
Dataflow mode	Weight-Stationary (WS)	Gemmini architecture

Table 4: Gemmini Hardware Parameters

4 Compiler Pipeline Reference

4.1 Lowering Passes

Pass	File Location	Role
ONNXToGemmini	Conversion/ONNXToGemmini/	Intercept supported ONNX ops; emit gemmini dialect or krnl.call
GemminiTiling	Transform/Gemmini/	Tile direct MatMul to DIM=16 blocks; assign ping-pong scratchpad slots
GemminiToGemminiLow	Transform/GemminiLow/	Convert high-level gemmini ops to instruction-level gemmini_low ops
GemminiLowRewrite	Transform/GemminiLow/	Peephole: eliminate redundant adjacent config/fence/memory ops
StaticScratchpadAllocation	Transform/GemminiLow/	Assign fixed scratchpad row offsets to each tile
GemminiLowToLLVM	Conversion/GemminiLowToLLVM/	LLVM/RISC-V CUSTOM_3 in-line assembly

Table 5: Gemmini Lowering Passes

4.2 Corrected RoCC Instruction Sequence

The 10-step sequence below is the exact order emitted by GemminiLowToLLVM.cpp for a 16x16 WS-mode direct MatMul. Note that FENCE is emitted before every MVOUT.

Step	Instruction	Key Operands
1	CONFIG_EX (funct=0)	rs1 = (acc_scale « 32) (1 « 16) dataflow_WS
2	CONFIG_MVIN_A (funct=0)	rs1 = (MVIN_SCALE_ID « 32) 0x01, rs2 = stride_bytes
3	MVIN_A (funct=2)	rs2 = (rows«48) (cols«32) spad_row_A
4	CONFIG_MVIN_B (funct=0)	Same as Step 2
5	MVIN_B (funct=2)	rs2 = (rows«48) (cols«32) spad_row_B
6	PRELOAD (funct=6)	rs2 = C_acc_addr 0x80000000
7	COMPUTE_PRELOADED (funct=4)	rs2 = 0xFFFFFFFF (no-bias sentinel)
8	FENCE	Memory and pipeline flush
9	CONFIG_MVOUT (funct=0)	rs1 = 0x02, rs2 = (acc_scale«32) stride_bytes
10	MVOUT (funct=3)	rs2 = (rows«48) (cols«32) acc_offset 0x80000000

Table 6: Corrected RoCC Instruction Sequence (10 steps with FENCE)

4.3 Cycle Counting Formula

The total cycle count for a 16x16 WS-mode direct MatMul is:

$$\begin{aligned}
 \text{Total Cycles} = & \underbrace{1}_{\text{CONFIG_EX}} + \underbrace{1}_{\text{CONFIG_MVIN_A}} + \underbrace{64}_{\text{MVIN_A}} + \underbrace{1}_{\text{CONFIG_MVIN_B}} + \underbrace{64}_{\text{MVIN_B}} \\
 & + \underbrace{1}_{\text{PRELOAD}} + \underbrace{256}_{\text{COMPUTE_PRELOADED}} + \underbrace{1}_{\text{FENCE}} + \underbrace{1}_{\text{CONFIG_MVOUT}} + \underbrace{64}_{\text{MVOUT}} \\
 = & 454 \text{ cycles}
 \end{aligned}$$

Seven Correctness Invariants

1. Use CUSTOM_3 (opcode 0x7b), not CUSTOM_0
2. CONFIG_MVIN required before each MVIN
3. PRELOAD required before COMPUTE_PRELOADED
4. rs2 = 0xFFFFFFFF in COMPUTE_PRELOADED (no-bias sentinel)
5. Bit 31 set in MVOUT rs2 (read from accumulator)
6. FENCE required before CONFIG_MVOUT and MVOUT
7. a_stride=1 (bit 16 of CONFIG_EX rs1), c_stride=1 (bit 48 of rs2)

5 Supported Operators

5.1 Complete Operator Table

ONNX Op	Path	Runtime Function / Constraints
MatMul int8	Direct RoCC	gemmini_low.* (no call); rank-2 static, 16x16 WS tiles
MatMul f32	Runtime	om_gemmini_matmul_f32_hw; rank-2 static; quantize to Gemmini to dequantize
MatMul f32 batched	Runtime	om_gemmini_matmul_f32_nd_hw; rank-N x rank-2
MatMulInteger	Runtime	om_gemmini_matmulinteger_i8i8acc32[_zp]; i8xi8 to i32 (integration test coverage)
QLinearConv	Runtime	om_gemmini_qlinearconv_i8[_bias]; rank-4 int8 NCHW; batch=1; group=1 (integration test coverage)
Conv f32	Runtime	om_gemmini_conv_f32[_bias]; rank-4 NCHW; dynamic batch; static C/H/W
ConvTranspose f32	Runtime	om_gemmini_convtranspose_f32[_bias]; col2im scatter-add; dynamic batch
Gemm f32	Runtime	om_gemmini_gemm_f32[_bias]; rank-2; transA/transB/alpha/beta
Relu f32	Runtime	om_gemmini_relu_f32; rank-4; dynamic batch
BatchNormalization f32	Runtime	om_gemmini_batchnorm_f32; rank-4 NCHW; inference only
Add f32	Runtime	om_gemmini_add_f32; rank-4 same-shape; dynamic batch
MaxPool f32	Runtime	om_gemmini_maxpool_f32; rank-4 NCHW; static kernel/stride/pads
AveragePool f32	Runtime	om_gemmini_avgpool_f32; rank-4 NCHW; static kernel/stride/pads
GlobalAveragePool f32	Runtime	om_gemmini_globalavgpool_f32; rank-4 NCHW
Softmax f32	Runtime	om_gemmini_softmax_f32; rank-2 or rank-4
Resize f32	Runtime	om_gemmini_resize_nearest_f32 / om_gemmini_resize_linear_f32
Pad f32	Runtime	om_gemmini_pad_constant/reflect/edge_f32; static spatial pads
Concat f32	Runtime	om_gemmini_concat_f32; 2-input rank-4 NCHW; any axis
Slice f32	Runtime	om_gemmini_slice_f32; rank-4; constant axes/starts/ends/steps
Transpose f32	Runtime	om_gemmini_transpose_f32_hw; rank-4; constant perm
Split f32	Runtime	om_gemmini_split_2/3/4_f32; rank-4; 2-4 outputs; any axis

ONNX Op	Path	Runtime Function / Constraints
Sigmoid f32	Runtime	om_gemmini_sigmoid_f32; rank-4; elementwise; dynamic batch
Mul f32	Runtime	om_gemmini_mul_f32; rank-4 same-shape; dynamic batch

6 Runtime Library Reference

6.1 Runtime Function Groups

Group	Count	Example Symbols
Integer / quantized	4	om_gemmini_matmulinteger_i8i8acc32[_zp], om_gemmini_qlinearconv_i8[_bias]
Float HW path	7	om_gemmini_matmul_f32_hw, om_gemmini_conv_f32[_bias], om_gemmini_gemm_f32[_bias]
Elementwise / norm	6	om_gemmini_relu_f32, om_gemmini_batchnorm_f32, om_gemmini_sigmoid_f32, om_gemmini_mul_f32
Pooling	3	om_gemmini_globalavgpool_f32, om_gemmini_maxpool_f32, om_gemmini_avgpool_f32
Spatial	11	om_gemmini_resize_nearest/linear_f32, om_gemmini_pad_constant/reflect/edge_f32, om_gemmini_concat_f32, om_gemmini_slice_f32, om_gemmini_split_2/3/4_f32, om_gemmini_transpose_f32_hw
Linear algebra	2	om_gemmini_gemm_f32, om_gemmini_gemm_f32_bias

Table 8: Runtime Function Groups (33 total public symbols)

6.2 OMTensor Data Structure

```

1 typedef struct OMTensor {
2     void *data;           // Pointer to tensor data (row-major)
3     int64_t *shape;       // Array of dimension sizes
4     int64_t *strides;     // Array of strides in elements
5     int64_t rank;         // Number of dimensions
6     int64_t numElements;  // Total elements (product of shape)
7     onnxDataType dataType; // ONNX data type enum
8 } OMTensor;

```

Listing 1: OMTensor Structure

7 Build System Reference

7.1 CMake Configuration Variables

Variable	Description
ONNX_MLIR_ENABLE_GEMMINI	Master switch (ON/OFF). Appends Gemmini to ONNX_MLIR_ACCELERATORS.
ONNX_MLIR_GEMMINI_MODE	Build mode: compile-only, hardware, or simulator
GEMMINI_ROOT	Path to Gemmini installation containing gemmini.h (default: ./gemmini)
RISCV_INSTALL	RISC-V toolchain and simulator prefix
RISCV_GCC	RISC-V cross-compiler command (default: riscv64-linux-gnu-gcc)

Table 9: CMake Configuration Variables

7.2 Build Commands

```

1 cmake -S . -B gemmini_toolchain_build \
2   -DONNX_MLIR_ENABLE_GEMMINI=ON \
3   -DONNX_MLIR_GEMMINI_MODE=compile-only \
4   -DONNX_MLIR_BUILD_TESTS=ON \
5   -DCMAKE_BUILD_TYPE=Release \
6   -G Ninja
7
8 cmake --build gemmini_toolchain_build -j"$(nproc)"

```

Listing 2: Compile-Only Build

```

1 cmake -S . -B gemmini_toolchain_build \
2   -DONNX_MLIR_ENABLE_GEMMINI=ON \
3   -DONNX_MLIR_GEMMINI_MODE=simulator \
4   -DRISCV_INSTALL="$HOME/riscv-gemmini" \
5   -DONNX_MLIR_BUILD_TESTS=ON \
6   -G Ninja

```

Listing 3: Simulator Build

7.3 Prerequisite Checking

```

1 # Check compile-only dependencies
2 scripts/check_prerequisites.sh --mode compile-only
3
4 # Check hardware toolchain
5 scripts/check_prerequisites.sh --mode hardware
6
7 # Check simulator installation
8 RISCV_INSTALL=/path/to/riscv ./scripts/check_prerequisites.sh --mode
  simulator

```

Listing 4: Prerequisite Check Examples

8 Testing Infrastructure

8.1 Test Hierarchy

Suite	Count	Command
MLIR lit tests	41	check-onnx_mlir-accelerators-gemmini
Integration tests	18	check-gemmini-integration
Example tests	22	bash gemmini/tests/run_all_tests.sh
Model zoo	9	run_model_zoo.py
Numerical correctness	3 models	simulator_numerical_correctness.py

Table 10: Test Suite Overview

8.2 MLIR Lit Test Categories

Category	Tests
MatMul	matmul_fp32.mlir, matmul_fp32_batched.mlir, matmul_i8.mlir, matmul_fp16.mlir
Direct RoCC	direct_matmul_rocc.mlir, direct_matmul_rocc_edge.mlir, direct_matmul_i8_rocc.mlir, config_transpose_bits.mlir
Conv / Gemm	conv2d_gemmini.mlir, conv2d_dynamic_batch.mlir, conv2d_dynamic_spatial.mlir, convtranspose_gemmini.mlir, gemm_gemmini.mlir, gemm_dynamic_batch.mlir
INT8 (integration coverage)	QLinearConv and MatMulInteger zero-point variants covered by integration tests (not lit)
Resize	resize_nearest.mlir, resize_linear.mlir
Pad	pad_constant.mlir, pad_reflect.mlir, pad_edge.mlir
Concat	concat_axis0.mlir, concat_axis1.mlir, concat_axis2.mlir, concat_dynamic_batch.mlir
Slice	slice_start_end.mlir, slice_negative_indices.mlir, slice_step.mlir, slice_dynamic_batch.mlir
Transpose	transpose_identity.mlir, transpose_nchw_to_nhwc.mlir, transpose_swap_hw.mlir, transpose_dynamic_batch.mlir
Split	split_axis1.mlir, split_axis2.mlir, split_dynamic_batch.mlir, split_3_outputs.mlir
Elementwise	sigmoid_static.mlir, sigmoid_dynamic_batch.mlir, mul_static.mlir, mul_dynamic_batch.mlir
Infrastructure	gemmini_low_rewrite.mlir, gemmini_double_buffer.mlir

Table 11: MLIR Lit Test Categories (41 total)

8.3 Example Test Tiers

Tier	Tests	Description
1	01, 11-14, 17, 19-transpose, 20-split, 21-sigmoid, 22-mul	f32 smoke; generate model to compile to check symbol
2	03, 04, 05, 07, 08	INT8: MatMulInteger (scalar/vector zp), QLinearConv, direct-RoCC MatMul
3	09, 10, 15, 16, 18	Multi-op networks; compile + link to RV64 ELF
4	06 (ResNet-18), 19-yolo-real (YOLOX-nano)	External models; SKIP gracefully when absent

Table 12: Example Test Tiers (22 total)

8.4 Model Zoo Results

Model	Category	Size	Gemmini ops	Result
ResNet-18 v1	Classification	44.7 MB	21	PASS
ResNet-50 v1	Classification	98 MB	54	PASS
MobileNetV2	Classification	13.3 MB	36	PASS
SqueezeNet 1.1	Classification	4.7 MB	64	PASS
DenseNet-121	Classification	32 MB	426	PASS
Res2Net-101	Classification	173 MB	467	PASS
YOLOX-nano	Detection	3.5 MB	332	PASS + Spike
YOLOv5s (FP32)	Detection	28 MB	201	PASS
BERT-tiny	NLP	43 MB	24	PASS

Table 13: Model Zoo Results (9/9 PASS)

9 Performance Characterization

9.1 RoCC Instruction Latency

Instruction	Latency (cycles)	Notes
CONFIG_EX	1	Configuration register write
CONFIG_MVIN	1	Load stride configuration
MVIN (per 16x16 tile)	64	16x16 tile, 4 banks
PRELOAD	1	Register setup
COMPUTE_PRELOADED (16x16)	256	$DIM^2 = 256$ MAC cycles
FENCE	1	Pipeline flush
CONFIG_MVOUT	1	Store stride configuration
MVOUT (per 16x16 tile)	64	Same as MVIN

Table 14: RoCC Instruction Latencies

9.2 PE Utilization Formula

$$\text{Utilization} = \frac{\text{actual MACs}}{DIM^2 \times \text{compute_cycles}} \times 100\% \quad (1)$$

Where:

- actual MACs = M x N x K (total multiply-accumulate operations)
- DIM = 16 (systolic array dimension)
- compute_cycles = number of cycles the systolic array is active
- Theoretical peak MACs/cycle = 256

9.3 Case Study: ResNet-18 Layer-by-Layer

Layer	M	N	K	PE Utilization
Conv1 (7x7)	64	112x112	3x7x7=147	88%
Conv2_1 (3x3)	64	56x56	64x3x3=576	94%
Conv2_2 (3x3)	64	56x56	64x3x3=576	94%
Conv3_1 (3x3)	128	28x28	128x3x3=1152	96%
Conv3_2 (3x3)	128	28x28	128x3x3=1152	96%
Conv4_1 (3x3)	256	14x14	256x3x3=2304	97%
Conv4_2 (3x3)	256	14x14	256x3x3=2304	97%
Conv5_1 (3x3)	512	7x7	512x3x3=4608	98%
Conv5_2 (3x3)	512	7x7	512x3x3=4608	98%

Table 15: ResNet-18 Convolutional Layers: PE Utilization Estimate

Cycle-Accurate Simulation

Cycle-accurate simulation of Gemmini requires Verilator or FPGA emulation. Spike provides only instruction-accurate simulation (no timing model). The latencies above are analytical estimates based on the Gemmini tile-cycle model.

10 Performance Benchmarks

10.1 Analytical MatMul Throughput

Hardware model: DIM=16, 1 GHz, 16 B/cycle scratchpad bandwidth.

Matrix Size	NumPy host (ms)	Gemmini hw est. (ms)	Speedup
128 x 128	3.038	0.014	2.09x
256 x 256	0.368	0.090	1.40x
512 x 512	1.629	0.623	1.46x
1024 x 1024	9.517	4.588	1.81x

Table 16: MatMul Throughput Benchmark

10.2 Double-Buffer DMA/Compute Overlap

Matrix Size	Tiles	Seq est. (ms)	Overlap est. (ms)	Benefit
32 x 32	8	0.000512	0.000384	25%
64 x 64	64	0.00256	0.001536	40%
128 x 128	512	0.01434	0.00819	43%
256 x 256	4096	0.09011	0.06554	27%

Table 17: Double-Buffer Overlap Benchmark

10.3 Numerical Correctness on Spike

Model	Max Abs Error	Max Rel Error	Pass
ResNet-18	1.097e-05	9.22e-04	YES
MobileNetV2	5.960e-06	7.99e-04	YES
BERT-tiny	3.338e-06	1.32e-03	YES

Table 18: Numerical Correctness at 1e-5 Tolerance

11 Usage Guide

11.1 Compiling a Model

```

1 ONNX_MLIR=gemmini_toolchain_build/Release/bin/onnx-mlir
2
3 # Emit MLIR for inspection
4 $ONNX_MLIR --maccel=Gemmini --EmitMLIR model.onnx -o /tmp/model_gemmini
5
6 # Count Gemmini runtime calls
7 grep -c 'om_gemmini_' /tmp/model_gemmini.onnx.mlir
8
```

```

9 # Compile to RV64 ELF (requires RISC-V toolchain)
10 $ONNX_MLIR --maccel=Gemmini \
11     --mtriple=riscv64-unknown-linux-gnu --mcpu=rocket \
12     model.onnx -o model_rv64
13
14 # Run on Spike
15 spike --extension=gemmini --isa=rv64imafdc "$HOME/riscv-gemmini/bin/pk"
    ./model_rv64

```

Listing 5: Basic Compilation Commands

11.2 Per-Operator Examples

Operator	Example Command	Expected Symbol
MatMul f32	bash gemmini/tests/...	om_gemmini_matmul_f32_hw
MatMulInteger	bash gemmini/tests/03- .../run_test.sh	om_gemmini_matmulinteger_i8i8acc32
QLinearConv	bash gemmini/tests/07- .../run_test.sh	om_gemmini_qlinearconv_i8_bias
Conv f32	bash gemmini/tests/...	om_gemmini_conv_f32_bias
Resize	bash gemmini/tests/12-resize- f32/run_test.sh	om_gemmini_resize_nearest_f32
Pad	bash gemmini/tests/13-pad- f32/run_test.sh	om_gemmini_pad_constant_f32
Concat	bash gemmini/tests/14-concat- f32/run_test.sh	om_gemmini_concat_f32
Slice	bash gemmini/tests/17-slice- f32/run_test.sh	om_gemmini_slice_f32
Transpose	bash gemmini/tests/19- transpose-f32/run_test.sh	om_gemmini_transpose_f32_hw
Split	bash gemmini/tests/20-split- f32/run_test.sh	om_gemmini_split_2_f32
Sigmoid	bash gemmini/tests/21- sigmoid-f32/run_test.sh	om_gemmini_sigmoid_f32
Mul	bash gemmini/tests/22-mul- f32/run_test.sh	om_gemmini_mul_f32

Table 19: Per-Operator Example Commands

11.3 Interpreting Emitted MLIR

What you see	Meaning
funcName = "om_gemmini_*" krnl.call in	Runtime-delegate path selected
gemmini_low.* ops	Direct-RoCC path selected (INT8 MatMul tiles)
llvm.inline_asm with CUS-TOM_3	Fully lowered RoCC instruction
Only onnx.* ops remain	Op fell through to standard scalar path

Table 20: Interpreting Emitted MLIR

12 Troubleshooting

12.1 Troubleshooting Decision Tree

Tree 1: “No Gemmini calls in output MLIR”

```

No Gemmini calls
  |
  v
Check build flags -> -DONNX_MLIR_ENABLE_GEMMINI=ON?
  |
  +-- NO -> Reconfigure with -DONNX_MLIR_ENABLE_GEMMINI=ON
  |
  +-- YES
      |
      v
Check ONNX ops -> Are they in the Supported Operators table?
  |
  +-- NO -> Op not yet implemented. Use fallback.
  |
  +-- YES
      |
      v
Check dynamic shapes -> Static shapes required for direct path
  |
  v
Run with --print-ir-after=onnx-to-gemmini

```

Tree 2: “Build fails with CMake errors”

```

CMake configuration fails
  |
  v
Run scripts/check_prerequisites.sh --mode <mode>
  |

```

```

+-- Missing gemmini.h -> Set -DGEMMINI_ROOT=/path/to/gemmini
|
+-- Missing RISC-V GCC -> Install gcc-riscv64-linux-gnu
|
+-- Missing Spike/pk/libgemmini.so
|
v
Run scripts/setup_gemmini_simulator.sh --install

```

Tree 3: “Spike illegal instruction”

```

Spike reports "illegal instruction"
|
v
Is --extension=gemmini present?
|
+-- NO -> Add --extension=gemmini to Spike command
|
+-- YES
|
v
Check CUSTOM_X opcode in disassembly
|
+-- CUSTOM_0 (0x0b) -> Wrong! Update to CUSTOM_3
|
+-- CUSTOM_3 (0x7b) -> Check XCUSTOM_ACC in gemmini_params.h

```

12.2 Debugging and Profiling Tools

12.2.1 MLIR Debugging Flags

```

1 # Print IR after every pass
2 onnx-mlir-opt --maccel=Gemmini --print-ir-after-all model.mlir
3
4 # Print IR before a specific pass
5 onnx-mlir-opt --maccel=Gemmini --print-ir-before=gemmini-tiling model.
  mlir
6
7 # Print IR after a specific pass
8 onnx-mlir-opt --maccel=Gemmini --print-ir-after=gemmini-low-to-llvm
  model.mlir

```

12.2.2 Runtime Verbosity

```

1 export OMP_DEBUG=1
2 export GEMMINI_DEBUG=1
3 export GEMMINI_VERBOSE=1

```

12.3 Common Issues Table

Issue	Solution
<code>-maccel=Gemmini</code> not recognized	Rebuild with <code>-DONNX_MLIR_ENABLE_GEMMINI=ON</code> .
Spike illegal instruction on <code>CUSTOM_0</code>	Add <code>-extension=gemmini</code> to Spike command; if the opcode in the binary is <code>CUSTOM_0</code> (0x0b), the compiler emitted the wrong channel — update to <code>CUSTOM_3</code> (0x7b). See Tree 3.
No Gemmini calls in emitted MLIR	Use <code>memory_access_checker.py</code> to see which ops are legal.
Numerical mismatch	Loosen tolerance to 1e-5 or compare with emulator.
Scratchpad range violation	Check tile placements within 16,384 rows (256 KB).
CMake cannot find Gemmini	Set <code>-DGEMMINI_ROOT=/path/to/gemmini</code> .
Simulator dependencies missing	Run <code>scripts/check_prerequisites.sh -mode simulator</code> and <code>setup</code> script.
BERT-tiny numerical mismatch	Use decomposed LayerNorm path or tolerance 1e-3.

13 Future Work Roadmap

1. **Hardware transposer for Conv:** Modify conv lowering to emit `config_ex` with OS mode and `transpose_a=true`, enabling hardware Im2Col.
2. **Im2Col pattern detection:** Add compiler pass that detects Im2Col patterns and replaces them with hardware transposer path.
3. **Dynamic shape support via CISC loops:** Expose `gemmini_loop_ws` and `gemmini_loop_conv_ws` primitives.
4. **Native FP16 support:** Extend `GemminiTargetInfo.hpp` to support `inputType = Float(5,10)`.
5. **DMA overlap profiling and tuning:** Profile workloads and tune queue depths and bus widths.
6. **Physical hardware validation:** Deploy to Chipyard SoC FPGA and re-run numerical correctness suite.
7. **Cycle-accurate profiling:** Use Verilator simulation or bare-metal Spike to access `rdcycle`.

14 References and Further Reading

14.1 Gemmini Repository

- URL: <https://github.com/ucb-bar/gemmini>

- Branch: master
- Hardware parameters: `gemmini/lib/gemmini-rocc-tests/include/gemmini_params.h`

14.2 ONNX-MLIR Repository

- URL: <https://github.com/onnx/onnx-mlir>
- Gemmini backend location: `src/Accelerators/Gemmini/`

14.3 Key Papers and Specifications

Reference	Citation
Gemmini (ASPLOS 2021)	Genc, H., et al. “Gemmini: Enabling Systematic Deep-Learning Architecture Evaluation via Full-Stack Integration.”
Systolic Arrays (Kung 1982)	Kung, H. T. “Why systolic architectures?” <i>Computer</i> , vol. 15, no. 1, pp. 37-46.
MLIR (CGO 2020)	Lattner, C., et al. “MLIR: A Compiler Infrastructure for the End of Moore’s Law.”
RISC-V RoCC Interface	RISC-V International, “RoCC Interface Specification,” 2019.
ONNX Specification	LF AI & Data Foundation, “ONNX Operator Specification v1.17.0,” 2026.

Table 22: Key References

15 Appendices

15.1 Appendix A: Environment Variables

Variable	Purpose
GEMMINI_ROOT	Path to Gemmini installation
RISCV_INSTALL	Path to RISC-V toolchain and simulator prefix
ONNX_MLIR_BUILD_DIR	Build directory for compiler binaries
LLVM_INSTALL	Path to LLVM installation
PK	Override proxy kernel path for Spike
SPIKE	Override Spike executable path
IN_N, IN_C, IN_H, IN_W	Input dimensions for model runners
OMP_DEBUG	Enable OMTensor debug prints
GEMMINI_DEBUG	Gemmini runtime debug output
GEMMINI_VERBOSE	Verbose logging

15.2 Appendix B: CMake Quick Reference

```

1 cmake -S . -B gemmini_toolchain_build \
2   -DONNX_MLIR_ENABLE_GEMMINI=ON \
3   -DONNX_MLIR_GEMMINI_MODE=compile-only \
4   -DONNX_MLIR_BUILD_TESTS=ON

```

Listing 6: Compile-Only Build

```

1 cmake -S . -B gemmini_toolchain_build \
2   -DONNX_MLIR_ENABLE_GEMMINI=ON \
3   -DONNX_MLIR_GEMMINI_MODE=simulator \
4   -DRISCV_INSTALL=/path/to/riscv-gemmini

```

Listing 7: Simulator Build

```

1 cmake -S . -B gemmini_toolchain_build \
2   -DONNX_MLIR_ENABLE_GEMMINI=ON \
3   -DONNX_MLIR_GEMMINI_MODE=hardware \
4   -DRISCV_GCC=riscv64-unknown-elf-gcc

```

Listing 8: Hardware Build

15.3 Appendix C: Runtime Function Summary

Function	Purpose
om_gemmini_tiled_matmul_i8i8acc32	Core-int8 MatMul (requires multiples of 16)
om_gemmini_quantized_matmul_f32	Quantize f32 to int8 matmul to dequantize
om_gemmini_matmul_f32_hw	FP32 MatMul
om_gemmini_matmul_f32_nd_hw	Batched FP32 MatMul (rank-N x rank-2)
om_gemmini_matmul_f16_hw	FP16 MatMul via f32 quantized path
om_gemmini_conv_f32, om_gemmini_gemm_f32	Conv/Gemm using quantized matmul path
om_gemmini_convtranspose_f32	ConvTranspose (col2im + quantized matmul)
om_gemmini_resize_nearest_f32, om_gemmini_resize_linear_f32	Resize (nearest/linear)
om_gemmini_pad_constant/reflect/edge_f32	Pad (constant/reflect/edge)
om_gemmini_concat_f32	Concat (2-input NCHW)
om_gemmini_slice_f32	Slice rank-4 with steps and negative indices
om_gemmini_transpose_f32_hw	Transpose rank-4 (hardware-preferred path)
om_gemmini_split_2/3/4_f32	Split 2-4 outputs
om_gemmini_sigmoid_f32	Sigmoid activation
om_gemmini_mul_f32	Elementwise multiplication

15.4 Appendix D: Glossary (Extended)

Term	Definition
DMA	Direct Memory Access — data movement between host memory and scratchpad

Scratchpad	On-chip SRAM managed by compiler; 16,384 rows x 16 bytes (256 KB)
Systolic Array	Pipelined compute array for matrix multiplication
Tile	16x16 block matching systolic array dimensions
Ping-Pong Buffer	Double-buffering technique for DMA/compute overlap
Weight Stationary (WS)	Dataflow where weights are kept in scratchpad across computations
Output Stationary (OS)	Dataflow where partial outputs are kept in accumulator
RoCC	RISC-V Custom Coprocessor interface (CUSTOM_3 opcode)
Spike	RISC-V architectural simulator
Proxy Kernel (pk)	Lightweight runtime for RISC-V baremetal simulation
Im2Col	Image to Column transformation for convolution
Col2im	Column to Image (scatter-add) for ConvTranspose
CISC Loop	Complex instruction loop for arbitrary matrix sizes
NCHW	Tensor layout: Batch (N), Channel (C), Height (H), Width (W)
LLVM IR	Low-Level Virtual Machine Intermediate Representation
MLIR	Multi-Level Intermediate Representation
PE	Processing Element — individual MAC unit in systolic array
Arithmetic intensity	Ratio of compute operations to memory accesses (FLOP/byte)
Bank conflict	Multiple memory requests targeting the same bank in the same cycle
Compute-bound	Performance limited by arithmetic throughput
Memory-bound	Performance limited by memory bandwidth
Prefetching	Loading data before it is needed
Gather DMA	DMA reading non-contiguous memory locations
Scatter DMA	DMA writing to non-contiguous memory locations
Strided DMA	DMA with non-unit stride between elements
Accumulator bypass	Direct data path skipping accumulator write
Hardware loop	CISC-style loop instruction without software overhead
Fence instruction	Synchronization primitive for memory ordering
Cache coherence	Protocol keeping caches consistent
Memory barrier	Instruction forcing memory operation ordering
In-flight DMA	DMA transfers issued but not completed
mvin stride	Bytes between rows in DMA load
mvout stride	Bytes between rows in DMA store
Config space	Configuration registers for RoCC commands
RoCC command queue	FIFO buffer for pending RoCC commands
Roofline model	Performance visualization of peak FLOP/s vs bandwidth

15.5 Appendix E: Validation Methodology

15.5.1 MLIR Lit Test Structure

```

1 // RUN: onnx-mlir-opt --maccel=Gemmini --convert-onnx-to-kernel --
   canonicalize %s | FileCheck %s
2
3 // CHECK-LABEL: func.func @test_conv
4 // CHECK: %alloc = memref.alloc() : memref<1x64x112x112xf32>
5 // CHECK: "kernel.call"(%alloc, %arg0, %arg1, %arg2) <{funcName = "
   om_gemmini_conv_f32_bias"}>
6 // CHECK-NOT: onnx.Conv

```

Listing 9: Example Lit Test

15.5.2 Numerical Tolerance Policy

Operation Type	Absolute Tolerance	Relative Tolerance	Reason
MatMul (f32 quantized)	1e-5	1e-3	Quantization noise
Conv (f32 quantized)	1e-5	1e-3	Quantization + im2col error
Elementwise (f32 scalar)	1e-6	1e-4	Pure scalar, bit-exact
Softmax (f32)	1e-5	1e-3	Exponential approximation
Resize (nearest)	0	0	Exact integer mapping
Resize (linear)	1e-5	1e-4	Floating interpolation

Table 26: Numerical Tolerance by Operation Type

15.5.3 CI/CD Pipeline Diagram

CI/CD Pipeline (GitHub Actions)

```

git push
|
v
Ubuntu 22.04 runner (32GB RAM, 8-core)
|
+---> Build (compile-only mode)
|
+---> MLIR lit tests (41 tests)
|
+---> Integration tests (18 tests)
|
+---> Example tests (22 tests)
|
+---> Model zoo (9 models)
|
v
All tests PASS?

```

YES	NO
v	v
Tag release	Fail build
(v3.0.0)	(email notify)

Validation Environment

Validation runs on Ubuntu 22.04 LTS with 32GB RAM, 8-core CPU, and 100GB disk space. Spike simulator is installed under /riscv-gemmini.

Revision History

Version	Date	Author	Changes
1.0	2026-05-21	SST Team	Initial reference manual
2.0	2026-05-22	SST Team	Added FP16, batched MatMul, simulator validation
2.1	2026-05-22	SST Team	Alignment status, test results, runtime verification
3.0	2026-06-08	SST Team	Comprehensive edition with Quick Start, Performance Characterization, troubleshooting, extended glossary
3.1	June 11, 2026	SST Team	CORRECTED: Hardware parameters (scratchpad rows: 16,384, 256 KB; accumulator rows: 1,024, 64 KB); Double-buffering slot addresses as relative ranges; RoCC sequence includes FENCE (10 steps, 454 cycles); Lit test table corrected (direct_matmul_i8_rocc.mlir, convtranspose_gemmini.mlir added; INT8 zero-point tests noted as integration coverage); Common Issues table updated (16,384 rows); Warning-box title corrected (Seven Correctness Invariants); CMake lit target name corrected (check-onnx_mlir-accelerators-gemmini); CUSTOM_0 troubleshooting entry updated with opcode fix guidance.

Prepared by: SST Team, SiliconeSignal Technologies

Location: Meknes, Morocco

Document Status: Final

Approval: Pending